Data Engineering [D.E]

Assignment-01

Name: A. Ascini

Branch: CSD

Roll-No:- 25NUSA4401

1) Compare and contrast data engineering with data science - How do they complement each other?

Data Engineering	Data Science
• Data engineering deals with designing, building and managing data pipelines and infrastructure	• Data science focuses on analyzing, data to extract insights and build predictive models
• To ensure reliable, clean and scalable data availability	• To derive meaningful insights & prediction from data
• Mostly raw and structured / unstructured data from multiple sources	• Mostly processed & cleaned data ready for analysis
• Data Ingestion, ETL, processes, data storage, data warehousing and pipeline maintainance	• Data analysis, Statistical modeling, ML and visualization
• SQL Hadoop, Spark, Kafka, Airflow cloud platforms.	• Python, R, Pandas, NumPy, Scikit-learn, Tensor Flow.
• Clean, Structured and accessible data sets.	• Reports, dashboards, Predictive model and insights.

How do they complement with each other:

→ Data Engineers prefer and organise data

→ Data Scientists analyze and generate insights from the data

→ without engineers, scientists lack quality data

→ without scientists, engineered data has limited business value.

Data Engineering

[D.E]

a) Explain why data engineering is considered the vital link in transforming raw data into business value?

Ans Data Engineering act as the vital link between raw data sources and business decision-making systems.

- Raw data generated from applications, sensors, logs and transactions is unstructured and cannot be used directly for business analysis.
- Data engineering enables data collection and integration from multiple heterogenous sources into a unified systems.
- It applies ETL processes to clean, validate and transform raw data into structured format.
- DE ensures that high data quality, accuracy and consistency which are essential for reliable insights.
- Scalable & automated data pipelines are built to handle large volumes & velocity of data efficiently
- Proper data storage architecture such as data warehousing and data lakes are designed for optimized access and security.
- DE supports real-time & batch data processing, enabling timely business decisions.
- Clean and well organized data produced by DE is used by analysts and data scientists for analytics & modeling.
- By making data reliable, accessible and timely, DE transforms raw data into ~~automatic~~ actionable business value.

Explain type A & type - B data engineers and data engineers inside a organization - internal facing v/s external facing data engineers.

Ans: Type - A Data Engineers: Abstract

- they build and maintain ETL pipelines and data warehouses.
- they work closely with analysts and business teams.
- their goal is to provide clean, accurate, analysis ready data.

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Type - B Data Engineers: [Build]

- Builds data tools and systems from scratch for special needs.
- they ensure scalability, performance & reliability of data platforms.

Data Engineers inside an organisation

Internal - facing data engineers:

- Internal facing data engineers build data systems for internal users.
- they support reporting, dashboards & internal analytics
- their work improves internal decision making & efficiency

External facing data engineers:

- External facing DE build data systems for external users & products.
- they design data API's & customer facing data services.
- their systems directly impact user experience & business revenue
- Provides data services, outside the organisation.

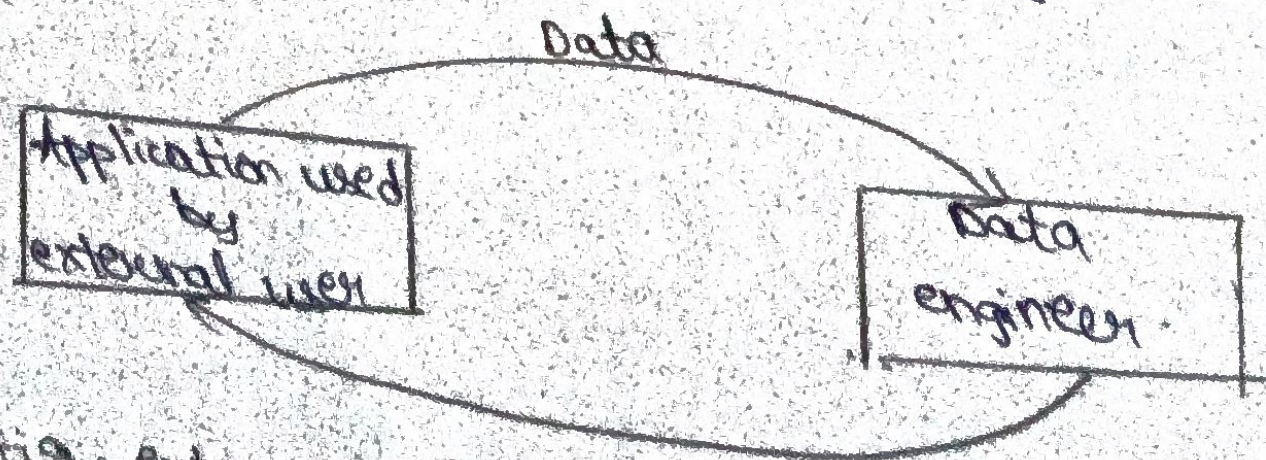

fig. external facing data engineer.

Engineering [n.E]

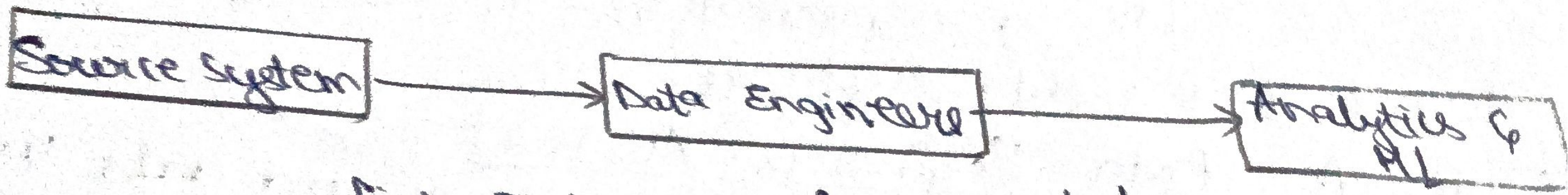

fig:- Internal facing data engineers